LLMs	Comparison with humans				Comparison with default goal		
	Spearman		MSE		vs. Both	vs. Inf.	vs. Social
	Original	Persona	Original	Persona			
GPT-4o	**0.75**	**0.76**	0.026	0.031	0.62	**0.99**	0.31
Claude-3.5-Sonnet	0.41	0.47	0.048	0.046	**0.73**	0.49	0.19
Llama-3.1-8B	0.11	0.15	0.052	0.052	0.77	**0.86**	0.71
Llama-3.1-8B-Instruct	0.17	0.17	0.061	0.063	**0.87**	0.75	0.78
Llama-3.1-70B	0.66	0.67	0.034	0.030	**0.86**	0.58	0.57
Llama-3.1-70B-Instruct	0.73	0.74	0.023	0.024	0.74	**0.75**	0.53
Llama-3.3-70B-Instruct	0.67	0.66	**0.018**	**0.019**	**0.80**	0.64	0.40
Mixtral-8x7B	0.36	0.35	0.043	0.044	0.74	**0.83**	0.19
Mixtral-8x7B-Instruct	0.43	0.39	0.080	0.082	**0.54**	0.41	0.10
Qwen2.5-72B	0.65	0.66	0.028	0.029	**0.83**	0.75	0.73
Qwen2.5-72B-Instruct	0.66	0.64	0.033	0.034	**0.63**	0.55	0.54

Table 1: Spearman correlations between the frequencies of human responses (Yoon et al., 2020) and LLM responses across all goal-rating combinations. Bold values indicate the highest correlation.

approximate the diversity in the population of human participants (Murthy et al., 2025; He-Yueya et al., 2024). For each model and prompting strategy, we sampled 30 responses with temperature $\tau = 1.0$ per scenario (see Appendix A for details of prompt design).

3.1 Model comparison

We report Spearman correlation and mean squared error (MSE) between LLM and human responses as an overall measure of fit (see Table 1). These results suggest that model size plays a crucial role in capturing human-like politeness strategies. Smaller models (`Llama-3.1-8B`) showed essentially no correlation with human responses, often failing to perform the multi-choice task at all, while intermediate-sized models like `Mixtral-8x7B` (effective model-size is 13B (Jiang et al., 2024)) showed only modest correlations. However, larger models ($\geq$70B parameters) demonstrated much stronger alignment with human behavior, with `Llama-3.3-70B-Instruct` achieving the highest correlations among open-source models (Spearman $r = 0.67$). Among closed-source models, `GPT-4o` displayed particularly strong performance (Spearman $r = 0.75$), while `Claude-3.5-Sonnet` lagged behind with more modest correlations ($r = 0.41$). These findings suggest that sophisticated pragmatic competence for politeness emerges primarily in larger models, potentially reflecting the greater contextual sensitivity needed to balance competing communicative goals.

3.2 Error analysis

Despite strong overall correlations (see Figure 1A), even the best-performing models showed systematic differences from human responses. To better understand these patterns, we conducted a detailed comparison with human responses following the visualization approach in Yoon et al. (2020). The results in Figure 1B show that the best-fitting open-source model captures many key features of the human response patterns. Most notably, when rating a poor performance (0/3 hearts) with both informational and social goals, the model appropriately deploys negation as a politeness strategy, just as humans do: both humans and LLMs prefer to say "wasn't terrible" rather than "was bad". The model also closely tracks human preferences for positive ratings (2-3 hearts), showing appropriate sensitivity to the quality of the performance.

However, key differences emerged in the granularity of responses. Where humans show graded preferences across response options (distributing probability mass across multiple choices), LLMs tend toward more categorical binary choices, either strongly preferring or completely avoiding certain responses. They consistently choose one single option given a context, rating, and goal combination in most cases—despite our efforts to increase

response diversity through temperature sampling ($\tau = 1.0$) or persona variation in prompting.

Closer analysis also revealed systematic differences in how well LLMs captured human behavior across different communicative goals. Models showed stronger alignment with human responses for the *social* goal but underperformed when the goal was to be purely *informative*. For example, when humans prioritize being informative about a poor performance, they often select direct negative feedback ("was bad"), while LLMs sometimes persist with softened language. This pattern suggests that, while LLMs have acquired some aspects of sophisticated politeness strategies, they may overapply these strategies even when directness would be more appropriate, potentially reflecting their training to be generally "helpful and harmless".

3.3 Default goal analysis

We included a *default* goal condition (no explicitly specified goal) to evaluate how LLMs respond without specific communicative instructions. This condition helps reveal the implicit goals that might have been induced through various stages of model training. Although the overall fit to human data varies across models, we can ask which explicit goal produces the closest response pattern to the default goal, as measured by Spearman correlation.

Overall, we find a stronger resemblance to the *both* goal (see Figure 1B), suggesting that models generally attempt to balance informativeness and social considerations by default. However, Table 1 reveals varying correlation patterns across different LLMs. While most models show stronger correlations with the *both* goal, others correlate more strongly with the *informative* goal. For instance, `Llama-3.3-70B-Instruct` appears to implicitly align with *both* (Spearman $r = 0.80$), whereas `GPT-4o` shows much stronger alignment with *informative* (Spearman $r = 0.99$).

These varied patterns suggest that the implicit goals guiding different LLMs' polite speech may reflect differences in their training objectives and alignment procedures. The dominant pattern of alignment with the *both* goal is consistent with the general instruction to models to be both helpful (informative) and harmless (socially appropriate). However, the variability across models indicates that while these systems have acquired sophisticated politeness capabilities, the specific ways they balance competing goals may differ from model to model.

4 Experiment 2: Open-ended Generation

While our multiple-choice experiment demonstrated that larger LLMs can reproduce basic patterns of goal-sensitive politeness strategies, such as the strategic use of negation, this constrained format limits our understanding of how models deploy politeness in naturalistic settings. In real-world interactions, speakers draw from a rich repertoire of linguistic devices beyond those provided in fixed-choice scenarios. This raises a critical question: how do LLMs perform when given the freedom to generate polite language from scratch?

To address this question, we designed an open-ended generation experiment that uses the same scenarios as our multiple-choice study but removes the response constraints. This approach allows us to examine whether LLMs employ a similarly diverse and context-sensitive set of politeness strategies as humans when both have access to the full expressivity of language, and directly compare to results in the constrained setting.

4.1 Methods

We used the scenarios from Yoon et al. (2020), preserving the same performance ratings (0-3 hearts) and communicative goals (informative, social, both, default). We collected 3 open-ended responses per scenario from 156 human participants via Prolific (each responding to 4 distinct scenarios) and three responses from LLMs that performed well in our first experiment: `GPT-4o`, `Claude-3.5-Sonnet`, and `Llama-3.3-70B-Instruct`. We selected these larger models after calibration experiments revealed that smaller models failed to generate coherent responses in the open-ended format (see Appendix B.6). Models were instructed to "keep responses short and concise" to ensure comparable length with human responses.

To assess preferences for these responses, we then conducted a two-alternative forced-choice evaluation with 156 human evaluators, each viewing four different scenarios. Evaluators made five judgments per scenario: (1) comparing human vs. LLM responses, (2-3) comparing goal-congruent vs. goal-incongruent responses for both sources, and (4-5) comparing rating-congruent vs. rating-incongruent responses for both sources. We randomized presentation order and ensured evaluators saw responses from different sources across blocks (see Appendix B for full details).

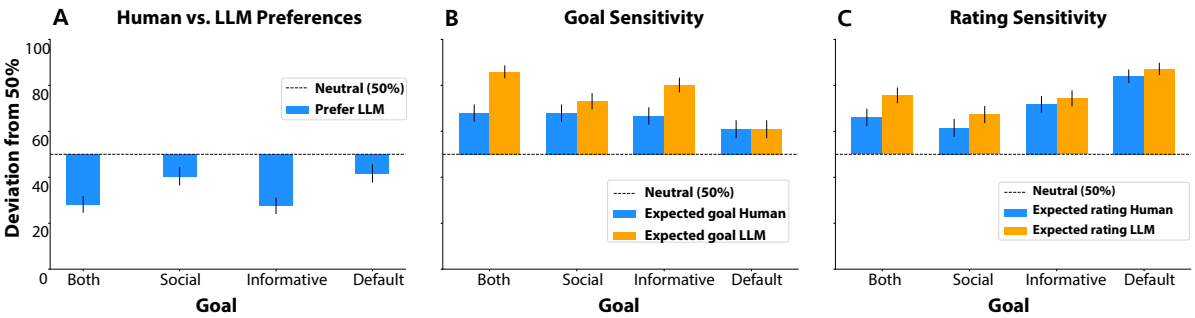

Figure 2: Human evaluation results. The bars show the relative preference (50% is chance). Bars above the 50% line indicate the percentage to which responses are preferred as expected, and below indicate the percentage to which responses are preferred as unexpected. (A) Evaluators systematically prefer LLM generations over human generations. (B) Both humans and LLMs are sensitive to goals and (C) ratings. Error bars are bootstrapped 95% confidence intervals.

4.2 Results

Overall preferences Surprisingly, human evaluators showed a marked preference for LLM-generated responses over human-generated ones across all goal types (66% of all trials; see Figure 2A). A mixed-effects logistic regression containing random intercepts at the evaluator and item level confirmed this preference was significantly different from chance ($z = 7.63$, $p < 0.001$). This pattern held for each of the four communicative goals, with the largest effect observed for the informative goal (22% above baseline) and the smallest effect observed for the default goal (8.3% above baseline; see Figure 2A). However, there were systematic differences in the strength of these preferences across goals; a model including a fixed effect of goal accounted for significantly more variance than the intercept-only model, according to a likelihood-ratio test $\chi^2(3) = 12.54, p = 0.006$.

Goal sensitivity Next, we considered the extent to which human-generated and LLM-generated utterances were goal-sensitive by calculating the proportion of trials where participants preferred a congruent utterance (i.e., an utterance actually produced to achieve the given goal) over an incongruent utterance (i.e., one produced under a different goal). We found that both humans and LLMs demonstrated sensitivity to communicative goals: evaluators preferred the goal-congruent human response 15.9% above-baseline ($z = 6.89, p < 0.001$), and preferred the goal-congruent LLM response even more strongly at 25.0% above baseline ($z = 9.59, p < 0.001$). Moreover, Figure 2B suggests that LLMs maintained greater or equal goal sensitivity across all four goals, indicating they successfully tailored their language to the specified communicative objective.

Rating sensitivity Finally, as a sanity-check, we asked whether utterances were sensitive to the actual state of the speaker (i.e., the number of hearts they felt about the performance being evaluated). Again, both groups showed strong rating sensitivity. Human responses achieved a 20.8% above-baseline preference for aligned ratings ($z = 8.32, p < 0.001$), while LLM responses demonstrated even higher sensitivity with a 26.1% above-baseline preference ($z = 9.59, p < 0.001$). As shown in Figure 2C, LLMs maintained equal or improved sensitivity across all goals, indicating that they are not simply producing generically polite utterances but are modulating their responses appropriately as a function of both the basic information to be conveyed (the rating) and the specified communicative goal (e.g., being informative vs. making someone feel good).

5 Linguistic Analysis of Politeness

While our evaluations show that LLMs successfully generate polite language that human evaluators prefer, these preferences alone don't reveal whether models use the same linguistic mechanisms as humans. To understand the specific politeness strategies employed by both humans and LLMs, we conducted a detailed linguistic analysis of the open-ended responses.

5.1 Negation

As a first step in our analysis, we examined how frequently the strategic use of negation documented in Yoon et al. (2020) and tested in Experiment 1